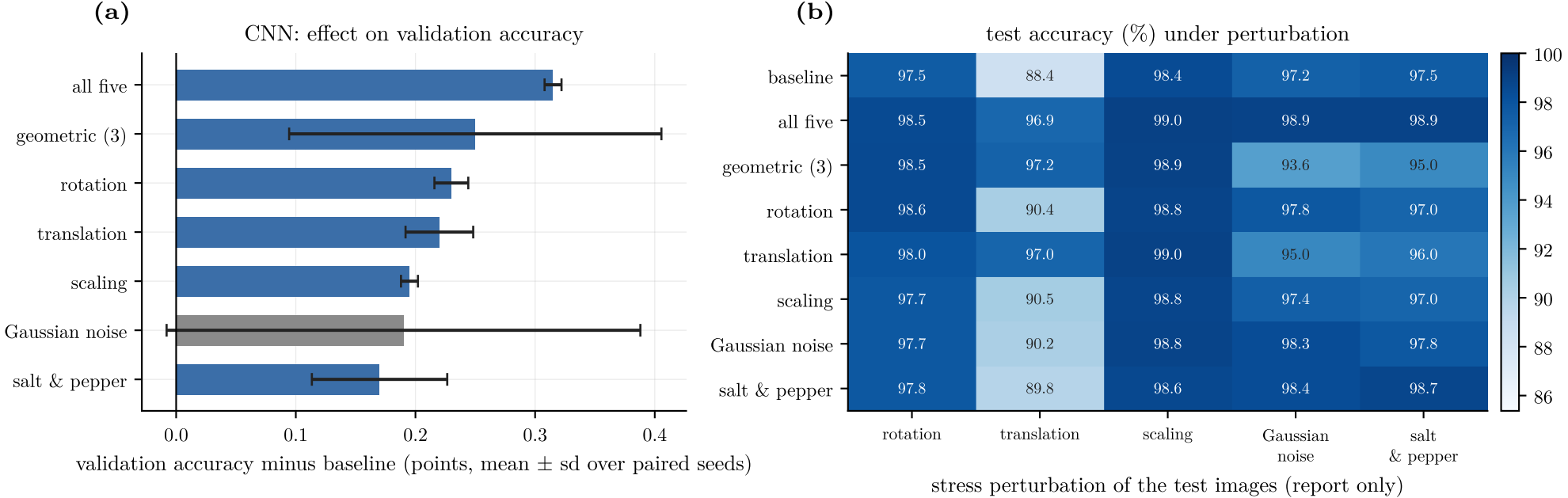
(a)
CNN: effect on validation accuracy
all five
geometric (3)
rotation
translation
scaling
Gaussian noise
salt & pepper
0.0
0.1
0.2
0.3
0.4
validation accuracy minus baseline (points, mean ± sd over paired seeds)
(b)
test accuracy (%) under perturbation
baseline
all five
geometric (3)
rotation
translation
scaling
Gaussian noise
salt & pepper
97.5 88.4 98.4 97.2 97.5
98.5 96.9 99.0 98.9 98.9
98.5 97.2 98.9 93.6 95.0
98.6 90.4 98.8 97.8 97.0
98.0 97.0 99.0 95.0 96.0
97.7 90.5 98.8 97.4 97.0
97.7 90.2 98.8 98.3 97.8
97.8 89.8 98.6 98.4 98.7
rotation
translation
scaling
Gaussian noise
salt & pepper
stress perturbation of the test images (report only)
100
98
96
94
92
90
88
86